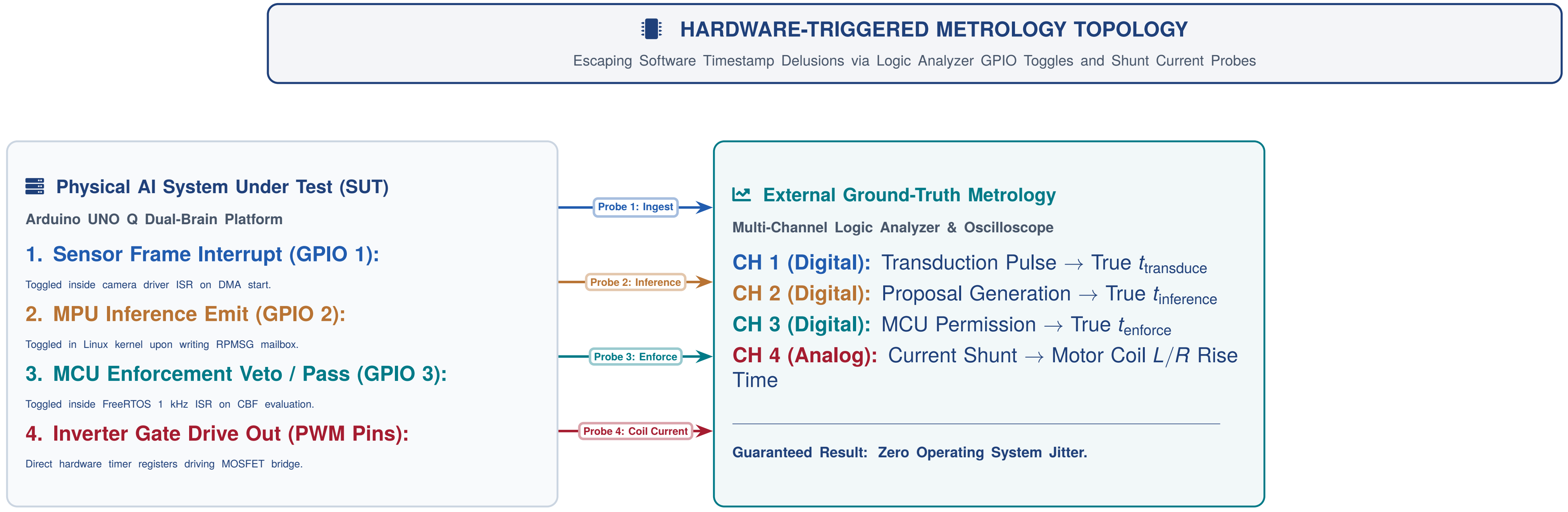
HARDWARE-TRIGGERED METROLOGY TOPOLOGY
Escaping Software Timestamp Delusions via Logic Analyzer GPIO Toggles and Shunt Current Probes
Physical AI System Under Test (SUT)
Arduino UNO Q Dual-Brain Platform
1. Sensor Frame Interrupt (GPIO 1):
Toggled inside camera driver ISR on DMA start.
2. MPU Inference Emit (GPIO 2):
Toggled in Linux kernel upon writing RPMSG mailbox.
3. MCU Enforcement Veto / Pass (GPIO 3):
Toggled inside FreeRTOS 1 kHz ISR on CBF evaluation.
4. Inverter Gate Drive Out (PWM Pins):
Direct hardware timer registers driving MOSFET bridge.
Probe 1: Ingest
Probe 2: Inference
Probe 3: Enforce
Probe 4: Coil Current
External Ground-Truth Metrology
Multi-Channel Logic Analyzer & Oscilloscope
CH 1 (Digital): Transduction Pulse $\rightarrow$ True $t_{transduce}$
CH 2 (Digital): Proposal Generation $\rightarrow$ True $t_{inference}$
CH 3 (Digital): MCU Permission $\rightarrow$ True $t_{enforce}$
CH 4 (Analog): Current Shunt $\rightarrow$ Motor Coil L/R Rise Time
Guaranteed Result: Zero Operating System Jitter.